Doctoral Candidate Thesis Declaration Form

You must complete the below declaration and include it in your thesis after your title page.

Failure to declare contributions may result in being referred to an Academic Misconduct Panel under our Plagiarism or Generative AI use policies.

If failure to declare is discovered after award, candidates may instead be referred to a Research Misconduct panel under our Research Integrity policy.

Part A: General Declaration

You must review and confirm all the points below.

☒ This thesis is a presentation of original work.

☒ This work has not previously been presented for a degree or other qualification at this University or elsewhere.

☒ All input or assistance in the creation of the academic work other than from the supervisory team (including AI) has been acknowledged in Part B below.

☒ Any text or data in the thesis that has been presented for publication (including in review) elsewhere is declared in Part B below.

Part B: Declaration of contribution from other sources

In this section, please declare any contributions from other sources under the headings in bold below. If not used, please write 'n/a' under the heading.

These declarations are intended to give context to the examiners and to ensure that doctoral researchers can use and acknowledge external support confidently.

Further guidance about what UCL considers 'other sources' and examples of completed Declaration Forms can be found in our policy on Transparency on Authorship and Generative AI in Doctoral Research.

Declarations

Data sources, if not collected yourself: include any paid/unpaid sources of data

SIRT patient data: Oxford University Hospitals NHS Foundation Trust - D. R. McGowan.

NEMA phantom with warm background: The Christie NHS Foundation Trust - J. Anton-Rodriguez.

NEMA phantom with cold background: National Physical Laboratory, Teddington - A. Fenwick, K. Ferreira and D. Deidda.

Anthropomorphic phantom: National Physical Laboratory, Teddington - K. Ferreira, A. Fenwick and D. Roddy (I assisted).

Partial volume correction test data: Matthias Koepp (UCL)

Software code, instrumentation development, etc : Describe support received to write code (from human or AI), build bespoke equipment, and any other shared work,

The reconstruction software was built on the open-source SIRF and STIR libraries, maintained by CCP SyneRBI (UKRI support: EPSRC grant EP/T026693/1), and CIL, maintained by CCPi (EPSRC grant EP/T026677/1).

SIMIND was used for scatter estimation: Michael Ljungberg (Lund University)

Claude (Anthropic) was used to assist in preparing research code for open-source publication.

AI: include any use of AI, including generative AI, not already included in the above categories where it has been used as a functional tool to assist in the process of creating the academic work.

Claude (Anthropic) was used to restructure and reformat text, and to improve the visual quality of plots and figures. ChatGPT (OpenAI) was used occasionally to bounce ideas off. All AI output was reviewed by me, and the scientific content, analysis and conclusions are my own.

Other: include any other material contributions to your thesis not already included in the above categories. Declare any contributions to proofreading other than standard editorial support, such as professional editing or substantial changes made by AI.

N/A

Publications: Include full references for all published work and work that has been submitted for publication that includes material shared with the thesis on which the candidate is an author.

- For each published work give a full citation including: all authors; title; journal/book (chapter); page numbers; volume; date of publication; publisher; DOI – if available.
- If manuscripts have been uploaded to a preprint server, please give full citations including DOI.
- Under each listed publication, please identify to which chapter of the thesis the publication relates.
- For multi-authored works, state your contribution and explain where this work appears in the final thesis.

S. Porter, D. Deidda, D. R. McGowan, J. Anton-Rodriguez, S. Arridge and K. Thielemans, “Weighted Directional Total Nuclear Variation for Joint Yttrium-90 PET/SPECT Reconstruction with CTAC-derived Guidance”, *arXiv preprint arXiv:2604.24437 [physics.med-ph]*, 27 April 2026. DOI: 10.48550/arXiv.2604.24437. Submitted to IEEE Transactions on Radiation and Plasma Medical Sciences; under review. **Thesis chapters:** Chapter 8 (adapted and extended from this preprint), with the prior definition in Chapter 6 and the optimisation approach in Chapter 7. **Contribution:** I developed the method, implemented the reconstruction software, designed and ran the phantom and clinical experiments, analysed the results and wrote the manuscript. D. Deidda, S. Arridge and K. Thielemans supervised the work and revised the manuscript. D. R. McGowan provided the SIRT patient data and clinical advice. J. Anton-

Rodriguez provided the warm-background NEMA phantom data and advice.

S. Porter, D. Deidda, K. Ferreira, A. Fenwick, D. Roddy, S. Arridge and K. Thielemans, “Synergistic PET-SPECT Reconstruction with Anatomically-Guided Regularisation”, in *2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD)*, IEEE, 2025, p. 1. DOI: 10.1109/NSS/MIC/RTSD57106.2025.11286725

Thesis chapters: Chapters 6 and 8. The results are superseded by the fuller analysis in Chapter 8.

Contribution: I developed the method, implemented and ran the reconstructions, analysed the results and wrote the abstract. K. Ferreira, A. Fenwick and D. Roddy provided the anthropomorphic phantom data from the National Physical Laboratory and helped acquire it. D. Deidda, S. Arridge and K. Thielemans supervised the work.

S. Porter, D. Deidda, D. McGowan, S. Arridge and K. Thielemans, “Simultaneous Multi-Bed MAP Reconstruction with CT-Guided Directional TV Prior for Y-90 PET SIRT”, *The PET is Wonderful Journal*, vol. 2, no. 1, October 2025. DOI: 10.2218/piwjournal.10841

Thesis chapters: Chapter 8, Section 8.3.1 (joint multi-bed PET reconstruction and the comparison with separately reconstructed and fused bed positions).

Contribution: I developed and implemented the joint multi-bed reconstruction, performed the reconstructions and comparison, and wrote the abstract. D. McGowan provided the SIRT patient data. D. Deidda, S. Arridge and K. Thielemans supervised the work.

S. D. Porter, D. Deidda, S. Arridge and K. Thielemans, “Optimising Subset Selection in Synergistic Emission Tomography Reconstruction”, in *2024 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and Room Temperature Semiconductor Detector Conference (RTSD)*, IEEE, October 2024, pp. 1–2. DOI: 10.1109/NSS/MIC/RTSD57108.2024.10654962

Thesis chapters: Chapter 7, which revisits these subset-selection experiments with a revised objective and extends them to prior-aware preconditioning.

Contribution: I implemented the algorithms, designed and ran the experiments, and wrote the abstract. D. Deidda, S. Arridge and K. Thielemans supervised the work.

S. Porter, K. Erlandsson, D. Deidda and K. Thielemans, “Regularised Richardson Lucy Deconvolution for PET PVC”, in *2025 IEEE Nuclear Science Symposium (NSS), Medical Imaging Conference (MIC) and*

Room Temperature Semiconductor Detector Conference (RTSD), IEEE, 2025, p. 1. DOI: 10.1109/NSS/MIC/RTSD57106.2025.11287560

Thesis chapters: Chapter 5 and Appendix A.3.

Contribution: I implemented the regularised deconvolution methods, performed the experiments and wrote the abstract. K. Erlandsson identified the gap in the literature that motivated this work, following his review of partial volume correction (H. Marquis, K. Erlandsson and I. Buvat, PET Clinics 20(4), 2025, DOI: 10.1016/j.cpet.2025.07.001), contributed through extensive discussion and provided some test data. M. Koepp provided the patient data. D. Deidda and K. Thielemans supervised the work.